

Sarvajanik Education Society

Shree Ramkrishna Institute of Computer Education and Applied Sciences

Behind P.T Science College, M.T.B. College Campus, Athwalines, Surat-395 001.

A Software Project On

Blood Bank & Donation Management System

as the partial fulfilment of

Semester -V study

for the degree of

Bachelor of Science (Computer Science)

(Affiliated to Veer Narmad South Gujarat University)

during the academic year 2022-2023.

Developed By:

Yagnik N. Devani

Guided By:

Dr.Priti Patel

Sarvajanik Education Society

Shree Ramkrishna Institute of Computer Education and Applied Sciences

Behind P.T. Science College, M.T.B College Campus, Athwalines, Surat-395 001.

CERTIFICATE

DEPARTMENT OF COMPUTER SCIENCE

This is to certify that

Student name	*Exam no*
1)Yagnik Devani	20192

has / have successfully completed his / her / their software engineering project work entitled

Blood Bank & Donation Management System

as the partial fulfilment of

Semester -V study

for the degree of

Bachelor of Science (Computer Science)

(Affiliated to Veer Narmad South Gujarat University)

during the academic year 2022-2023.

_____	_____
Internal Project Guide	**HOD**
	Dept. of Computer Science
Date: _____	**Place: Surat**

University Examination:

Examination No. : _____

Signature of External Examiners: _____

_____

INDEX for Project Documentation

SR.NO	CONTENTS	Page. No
1.	**Introduction**	4
	• Project Profile	
2.	**System Introduction**	5
	• System Definition, Objective, and scope	
	• Hardware & Software Requirements	
3.	**Requirement Analysis & Modeling**	7
	• Expected working of the system	
	• Data Flow Diagrams (DFD)	
	• Process Specification	
	• Data Dictionary	
4.	**Design**	20
	• ER Diagram & Database Design	
	• System Architecture	
	• System Flowchart	
	• Form & Report Designs (Screenshots) Layouts	
5.	**Coding**	32
	• Coding Approach/style	
	• Code Snippets	
6.	**Testing**	39
	• Testing Data Inputs (Data Validations)	
	• Testing Operations (Actions on each screen)	
7.	**New Tools/ Technologies learned/used**	44
8.	**System Limitations/Restrictions and Dependencies/Constraints**	45
9.	**Future Enhancement & Opportunities**	46
10.	**Bibliography & References**	47

1. INTRODUCTION

1.1 System Profile

- Project title : Blood & Bank Donation Management System
- Front-end : HTML , CSS , JS
- Back-end : PHP , MySQL
- Submitted By : Devani Yagnik N
- Project guided : By Dr. Priti Patel
- Organization : Shree Ramkrishna Institute of Computer Education & Applied Science

2. SYSTEM INTRODUCTION

2.1 System Definition, Objective and scope

2.1.1 System Information: -

- is designed in such a manner that it can suit the needs of all the blood bank requirements in The project blood bank management system is known to be a pilot project that is designed for the blood bank to gather blood from various sources and distribute it to needy people who have high requirements for it.

- The Software is designed to handle the daily transaction of the blood bank and Search the details when required.

- It also helps to register the details of donors, blood collection details, as well as blood, issued reports

- The software Application the course of the future.

- It will help us to find the Blood group with its most efficient time to take care of the blood and it is easier to hand over the blood to the hospital to help people to get blood on time.

- This all things have been stored and seen in this blood bank management system. To help more people try their best to do so.

2.1.2 Project Objective: -

- The main objective of the Blood Bank Management System is to manage the details of Blood, Donor, and Blood Group.
- It manages all the information about Blood.
- project is built at the administrative end and thus only the administrator is guaranteed the access
- Admin and user can operate the system 24/7.
- To check the availability of blood bags anytime
- To manage the information of its blood donor
- Support fast searching to find match blood bags for the right person.

2.1.3 System Scope: -

- Login function accessed by only admin .
- Admin can only delete user.

2.2 Hardware & Software Requirements

Hardware:-

- Processor : Intel core i3 or above
- Ram : 2 GB or above
- Hard-Disk : 100 GB or above

Software:-

- Backend : PHP & My Sql (PHP MY Admin).
- Frontend : HTML , CSS , JS
- Operating System : Windows 8 or above

3. REQUIREMENT ANALYSIS

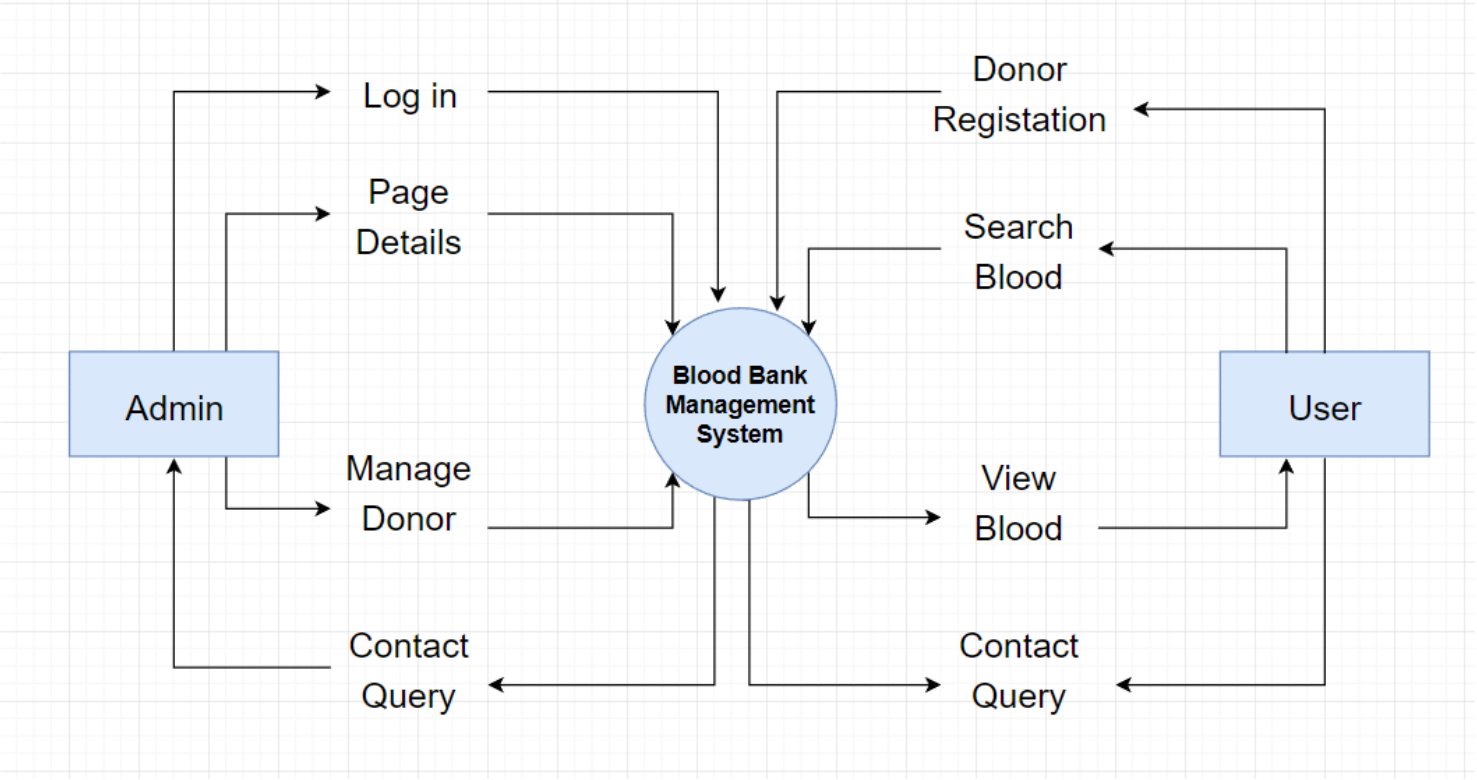
3.1 Expected working of System

- **Admin Module :** Admin can login with username and password which is already store in System database.
  - Login Facility
  - Donor Manage
  - Check Contact Query
  - Camp Manage
  - Pages Manages
  - Change Password
  - Logout

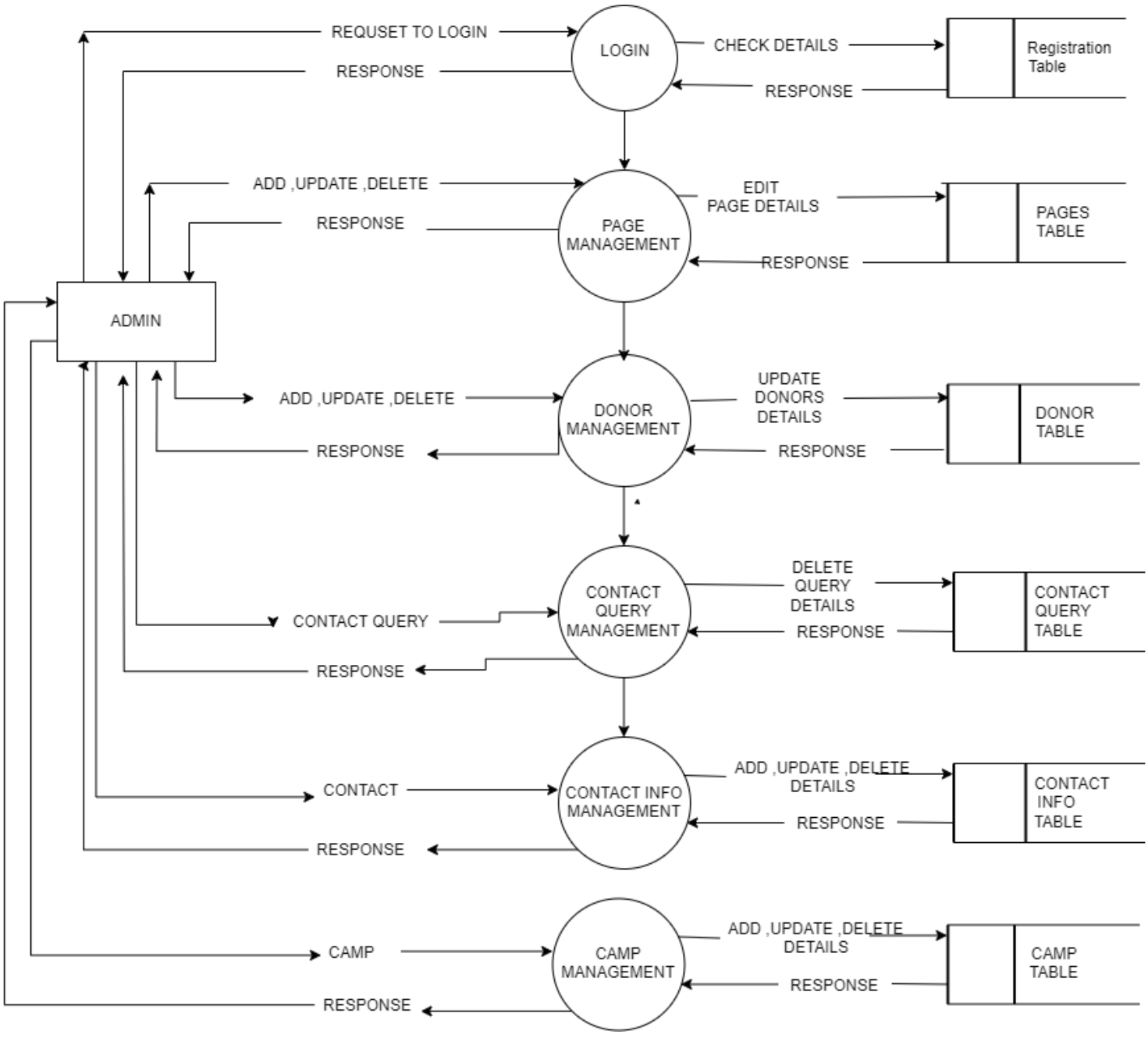
- **User Module :** User can donate blood and search blood donor as per blood group.
  - View Pages
  - View Camp Details
  - View Blood Donor Info
  - Registration as Donor
  - Search Available Blood
  - Contact Admin

3.2 Data Flow Diagrams (DFD)

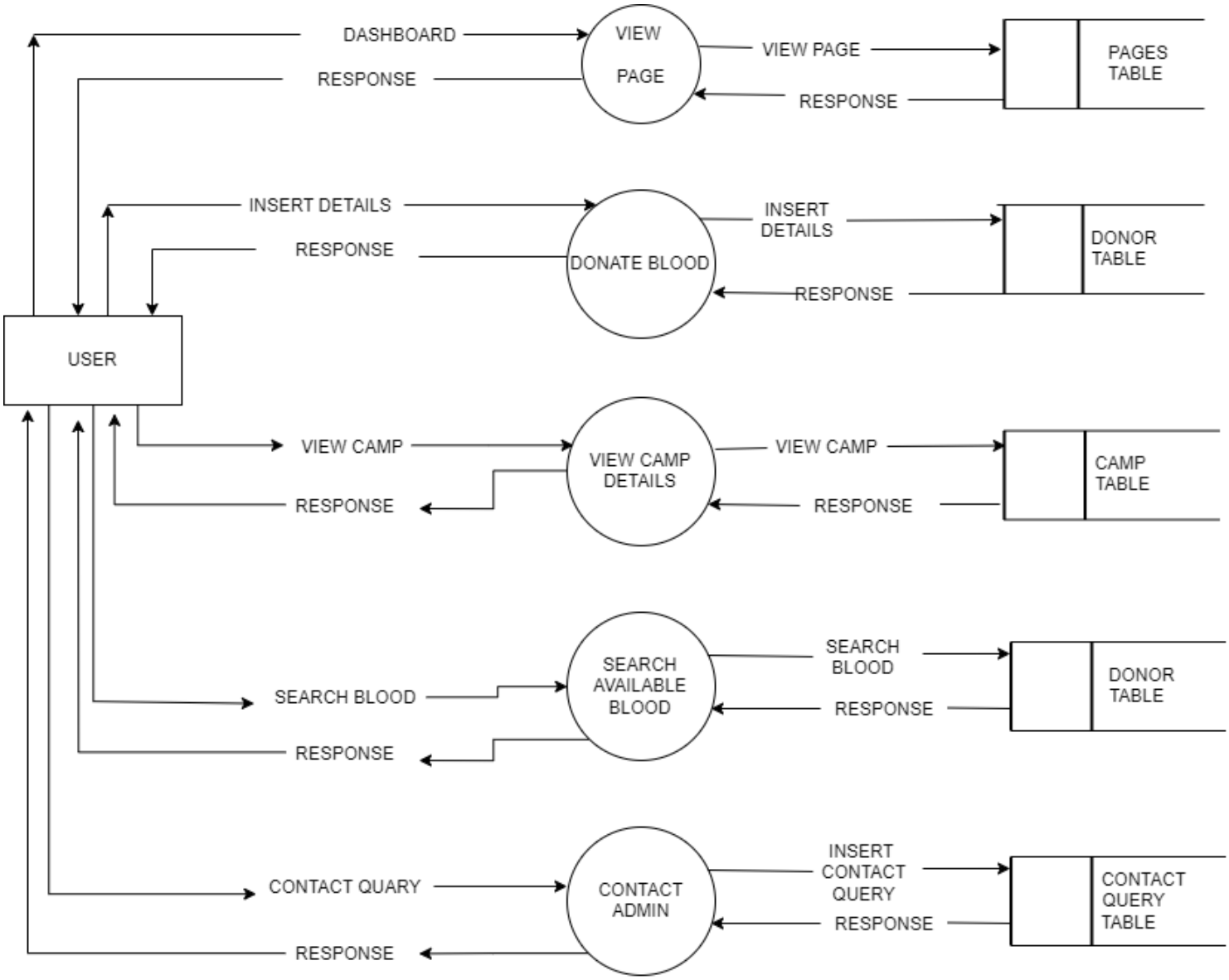
0 LEVEL DFD:-



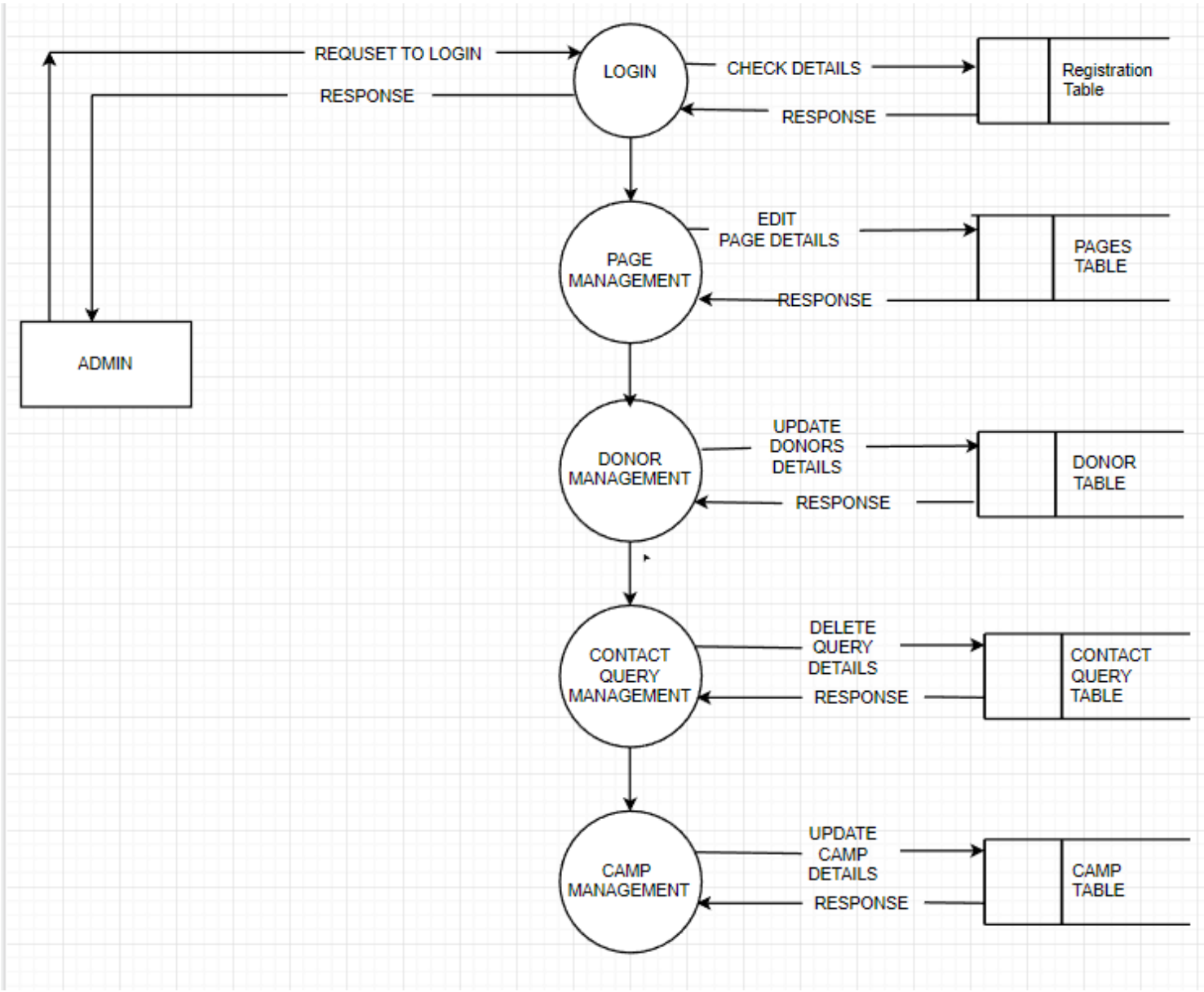
- **1st LEVEL DFD FOR ADMIN :-**



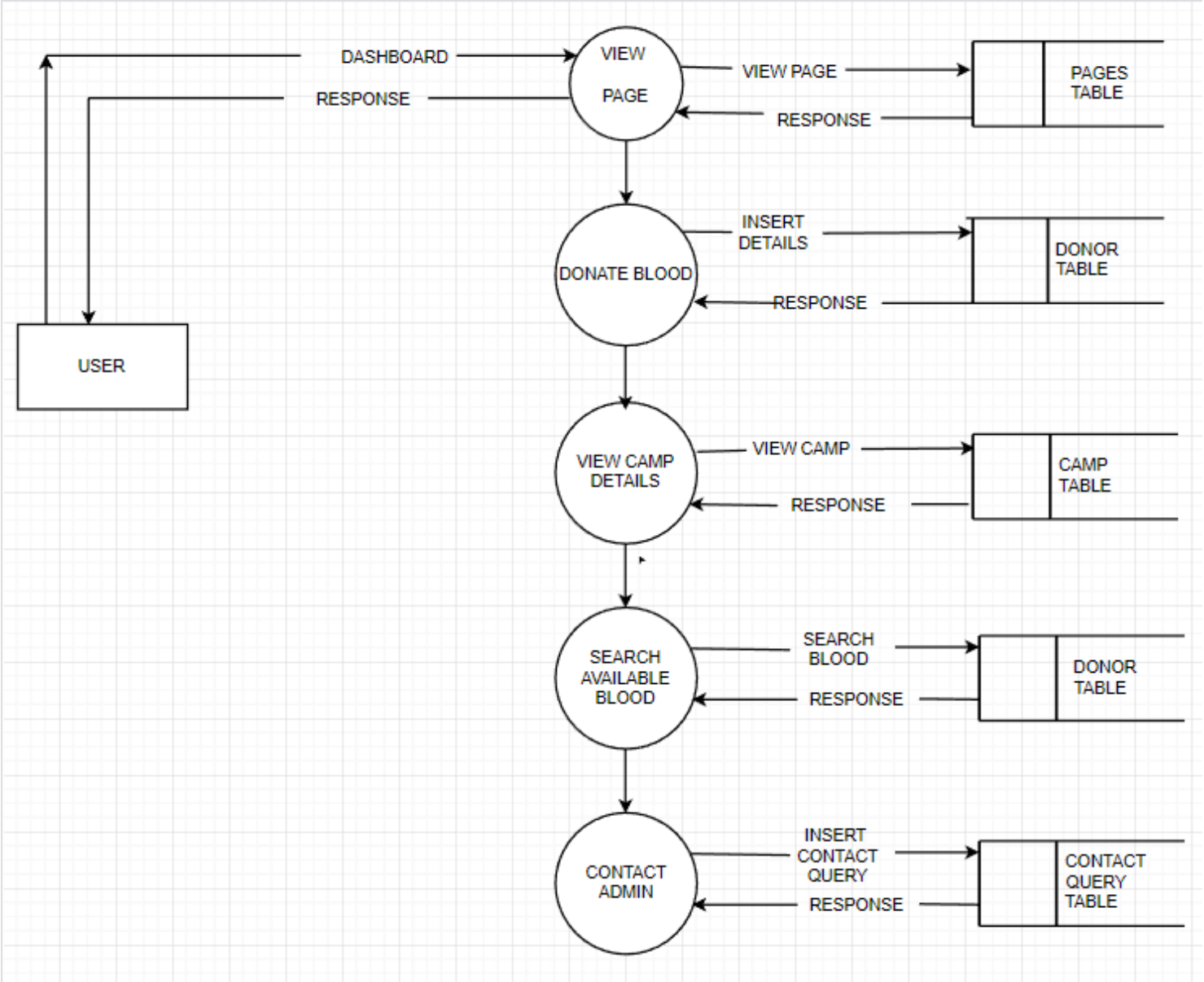
- **1st LEVEL DFD FOR ADMIN :-**


DASHBOARD
RESPONSE
VIEW
PAGE
VIEW PAGE
RESPONSE
PAGES
TABLE
INSERT DETAILS
RESPONSE
DONATE BLOOD
INSERT
DETAILS
RESPONSE
DONOR
TABLE
USER
VIEW CAMP
RESPONSE
VIEW CAMP
DETAILS
VIEW CAMP
RESPONSE
CAMP
TABLE
SEARCH BLOOD
RESPONSE
SEARCH
AVAILABLE
BLOOD
SEARCH
BLOOD
RESPONSE
DONOR
TABLE
CONTACT QUARY
RESPONSE
CONTACT
ADMIN
INSERT
CONTACT
QUERY
RESPONSE
CONTACT
QUERY
TABLE

- **2nd LEVEL DFD FOR ADMIN :-**



- **2ST LEVEL DFD FOR USER :-**



3.3 Process Specification

▶ PROCESS 1.0:

Name:	**Login**
Description:	This process is save all the information of the trust manager and Admin like Username , Password etc.

▶ PROCESS 1.0 FOR DONOR:

Name:	**Registration**
Description:	This process is save all the information of the trust manager or camp manager and donor like Username , Password etc.

▶ PROCESS 2.0 FOR ADMIN:

Name	**Manage User datails**
Description:	Admin can update and delete donor details

▶ PROCESS 3.0 FOR ADMIN:

Name:	**Manage Subject**
Description:	Admin can add camp and also delete camp

▶ PROCESS 4.0 FOR Donor:

Name:	**Fill query form**
Description:	Donor can fill the query form.

▶ PROCESS 5.0 FOR User:

Name:	**Search Available Blood**
Description:	Users can search for available blood

▶ PROCESS 6.0 Blood Donation:

NAME:	Blood Donation
Description:-	New donor ADD DATA Internet

3.4 Data Dictionary

- **admin_info Table**

Field Name	Data Type	Size	Constraint
Admin_id	int	10	Primary Key, Auto increment
Admin_name	text	50	Not Null
Admin_username	varchar	50	Not Null
Admin_password	varchar	50	Not Null

- **blood Table**

Field Name	Data Type	Size	Constraint
blood_id	int	11	Primary Key, Auto increment
blood_group	text	10	Not Null

- **camp Table**

Field Name	Data Type	Size	Constraint
camp_id	int	10	Primary Key, Auto increment
camp_name	text	50	Not Null
Date	Date	50	Not Null
Address	Varchar	50	Not Null
Contact	int	10	Not Null
Conductedby	text	50	Not Null
Organizedby	text	50	Not Null
time	varchar	50	Not Null

- **contact_info Table**

Field Name	Data Type	Size	Constraint
contact_id	int	10	Primary Key, Auto increment
contact_address	varchar	50	Not Null
contact_mail	Varchar	50	Not Null
contact_phone	int	10	Not Null

- **pages Table**

Field Name	Data Type	Size	Constraint
page_id	int	10	Primary Key, Auto increment
page_name	varchar	255	Not Null
Page_type	Varchar	50	Not Null
Page_data	longtext	-	Not Null

- **donor_details Table**

Field Name	Data Type	Size	Constraint
donor_id	int	10	Primary Key, Auto increment
donor_name	text	50	Not Null
Donor_number	Int	10	Not Null
Donor_mail	Varchar	50	Not Null
Donor_age	int	3	Not Null
Donor_gender	varchar	10	Not Null
Donor_blood	varchar	10	Not Null
Donor_address	varchar	100	Not Null

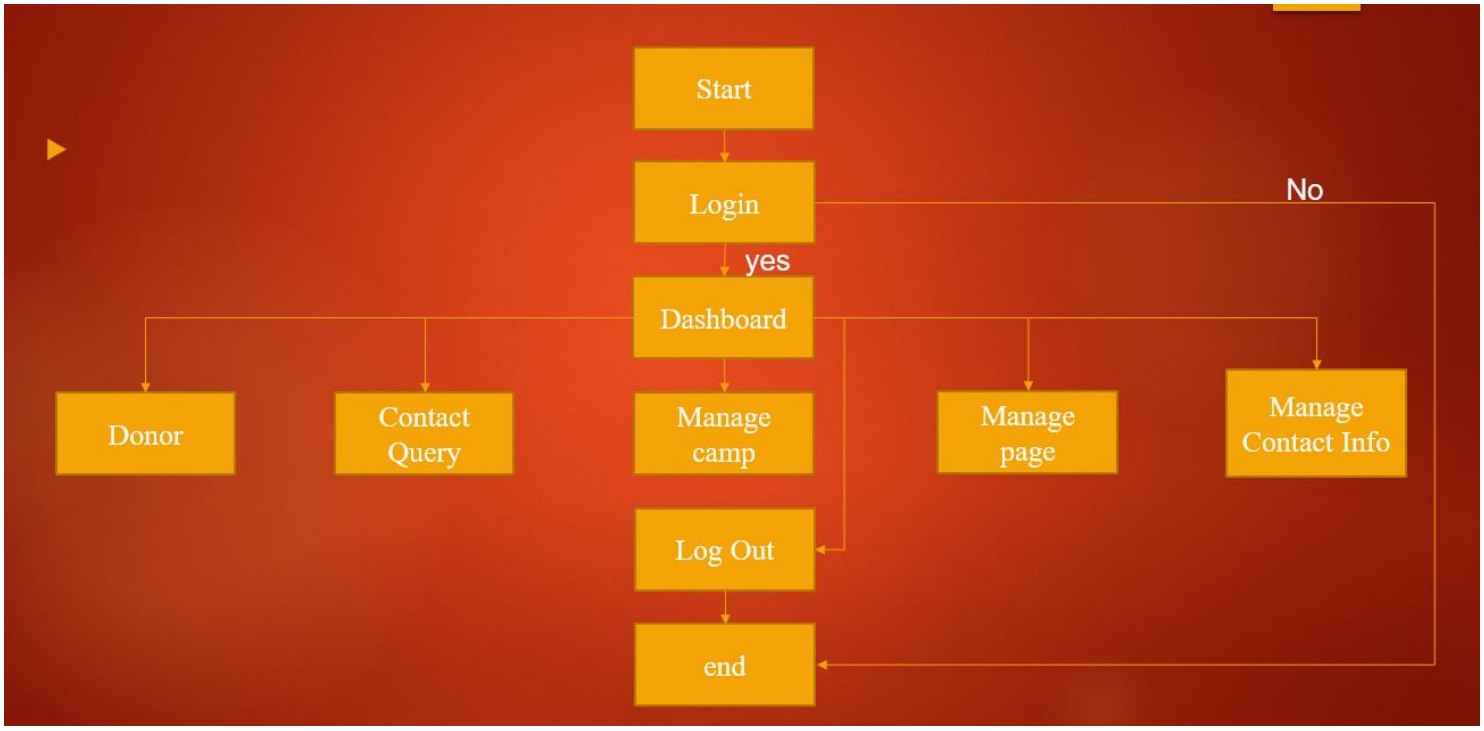
- **contact_query Table**

Field Name	Data Type	Size	Constraint
query_id	Int	10	Primary Key, Auto increment
query_name	text	50	Not Null
query_mail	Varchar	100	Not Null
query_number	int	10	Not Null
query_message	longtext	-	Not Null
date	timestamp	50	Not Null

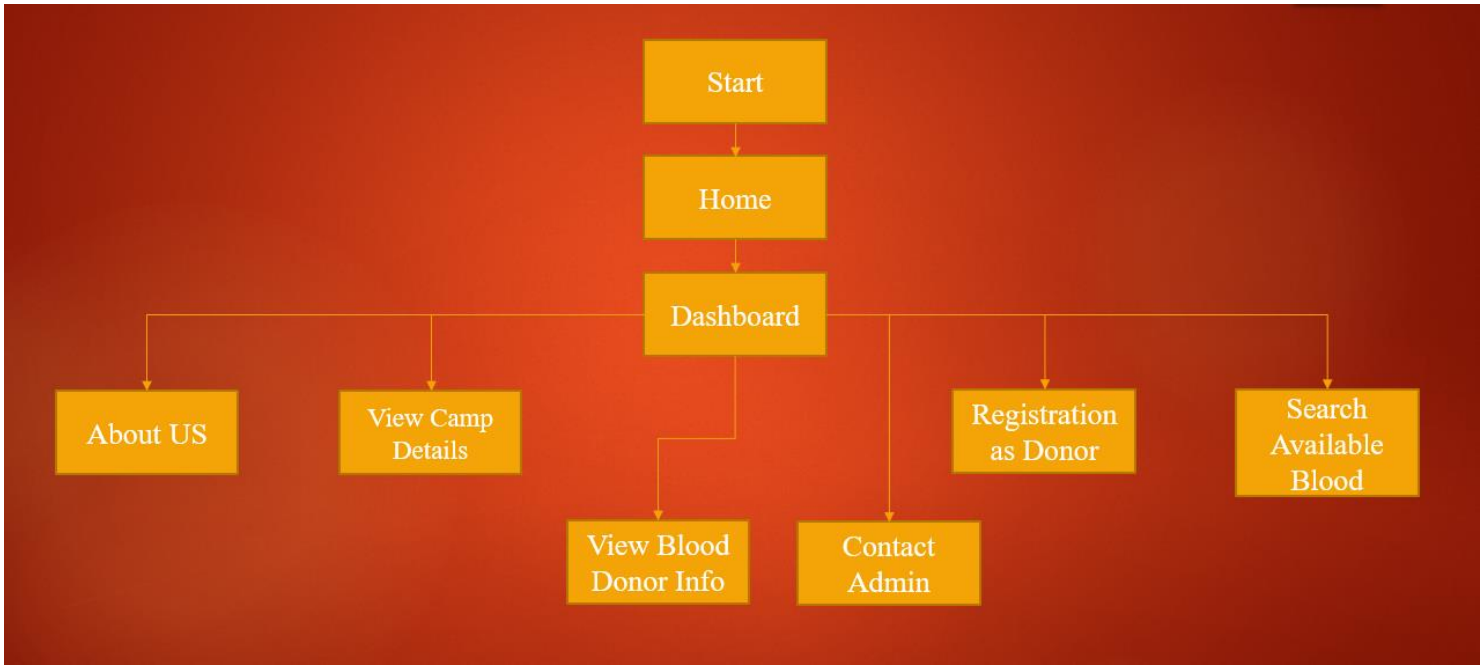
4. Design

4.1 System Flowchart :

4.1.1 Admin Side :



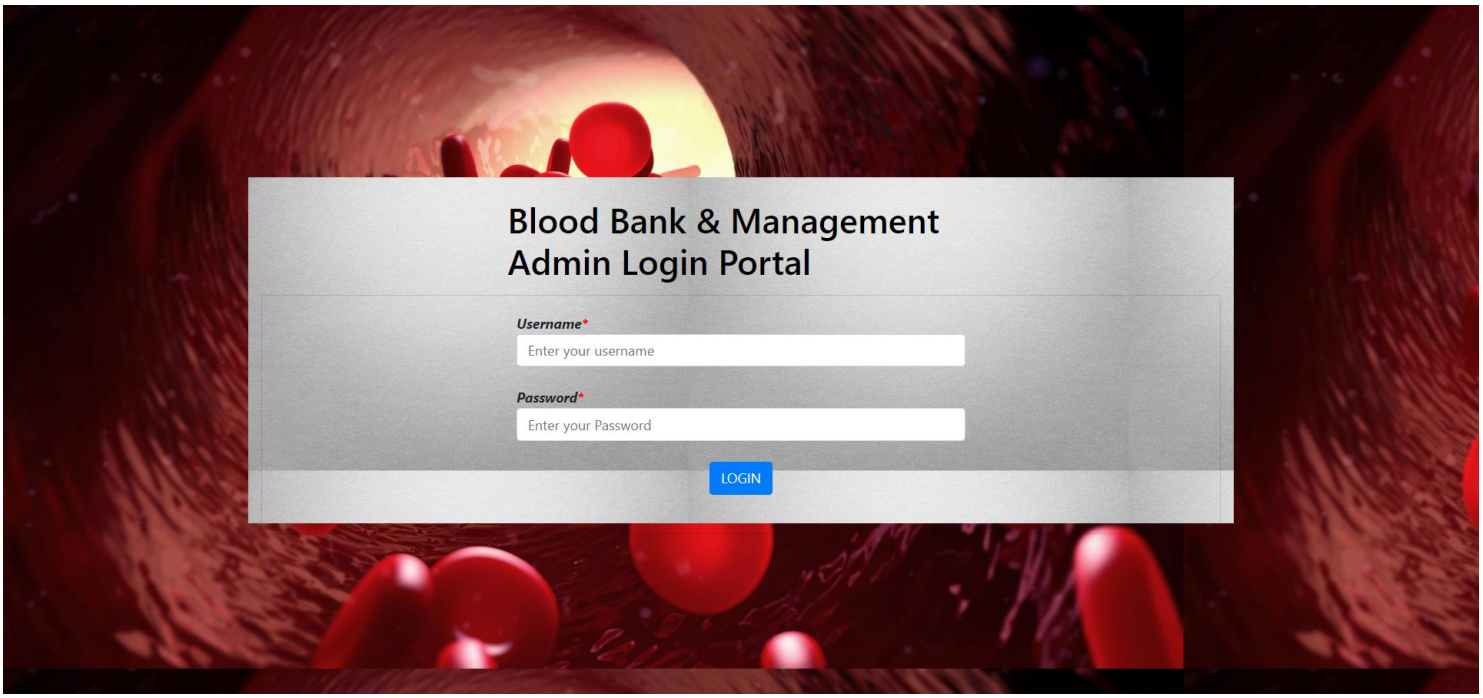
4.1.1 User Side :



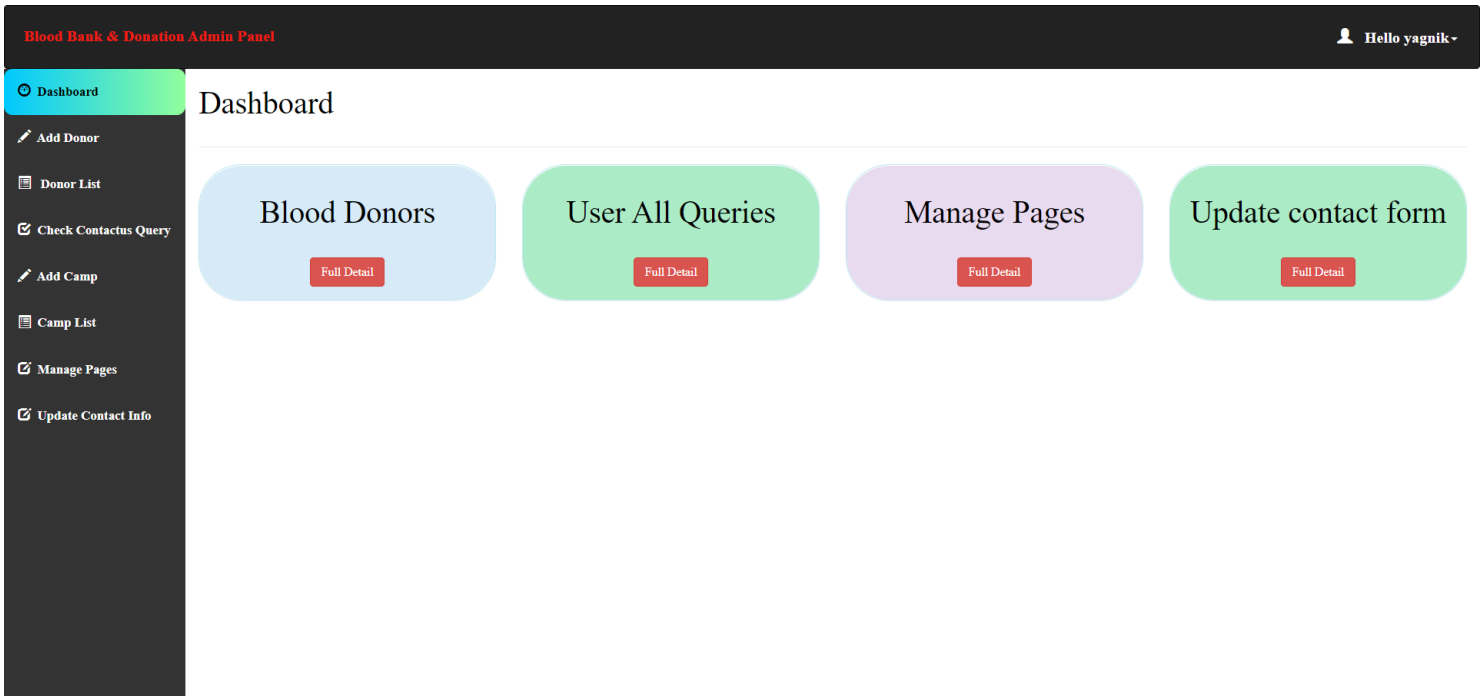
4.4 FORM AND REPORT DESIGN(SCREENSHOT)LAYOUT:

Admin Side :

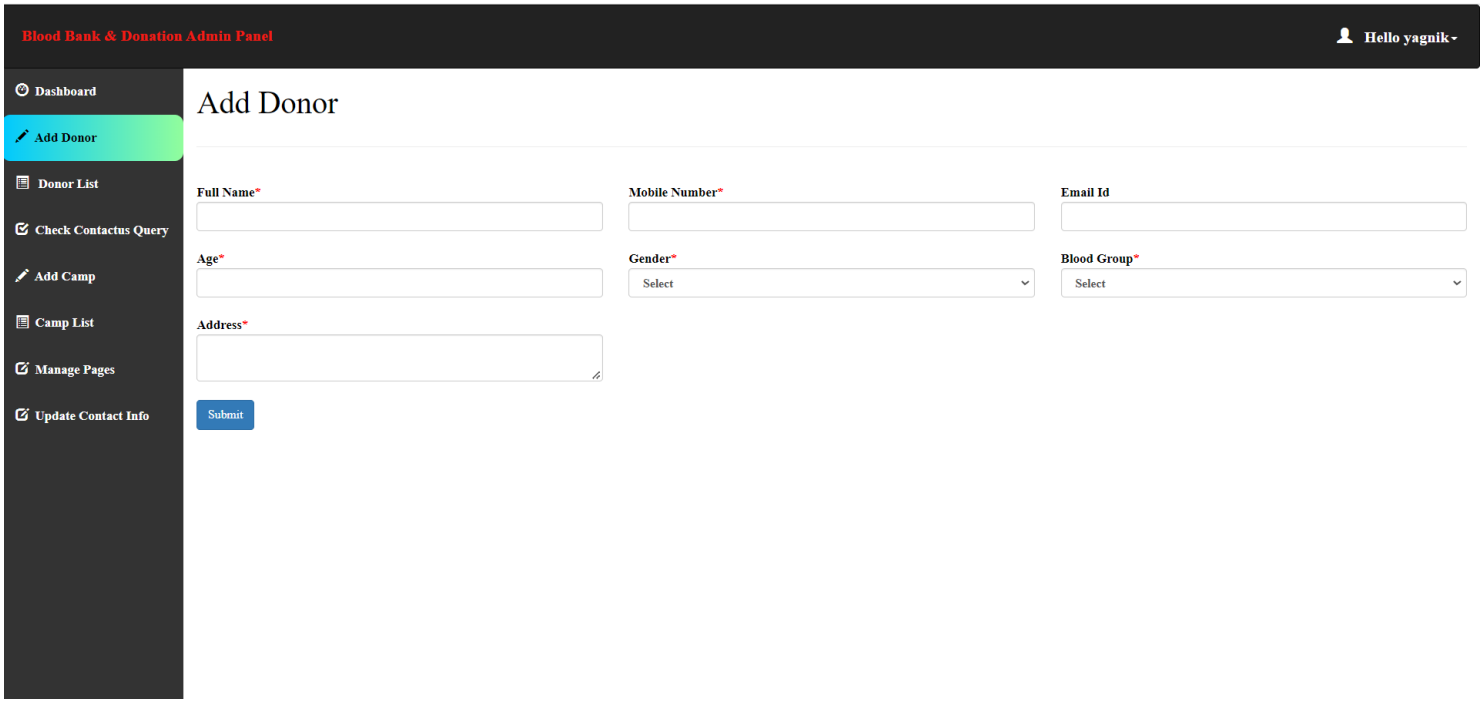
- **Login Page**



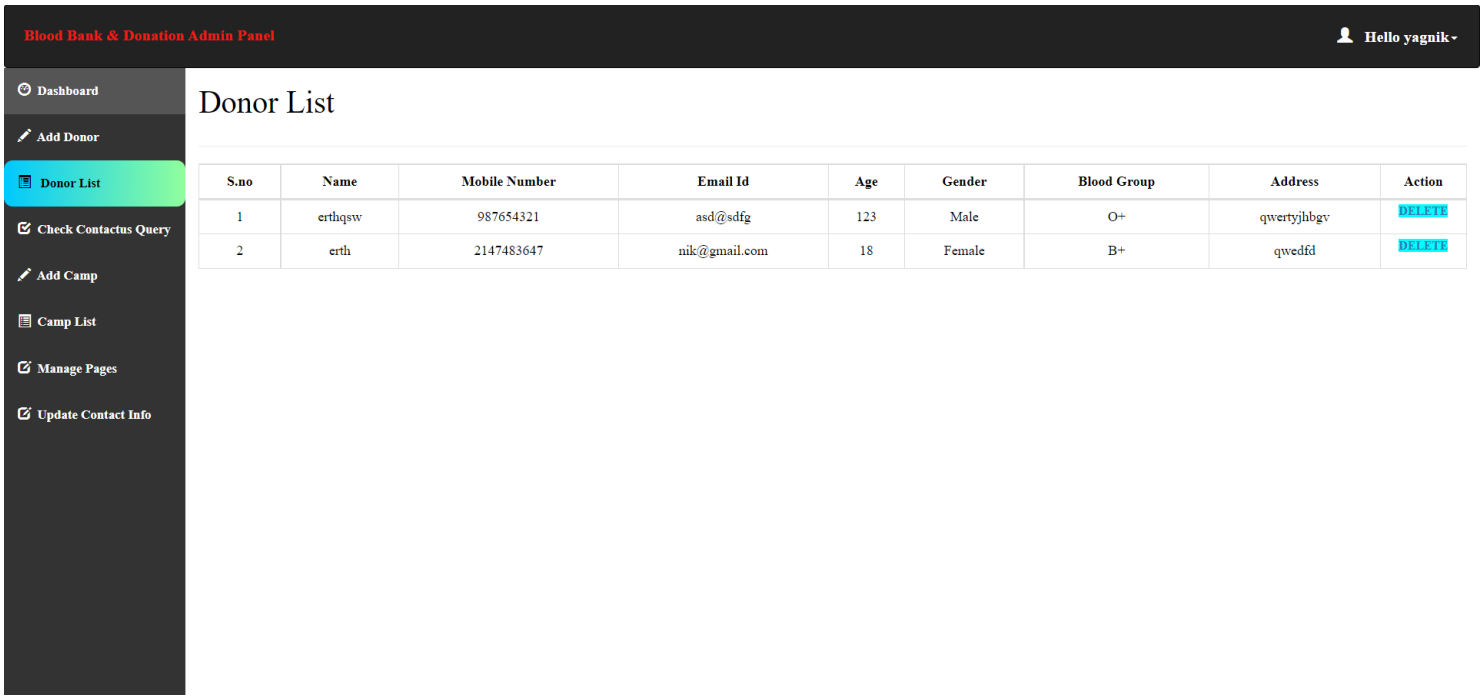
- **Dashboard Page**



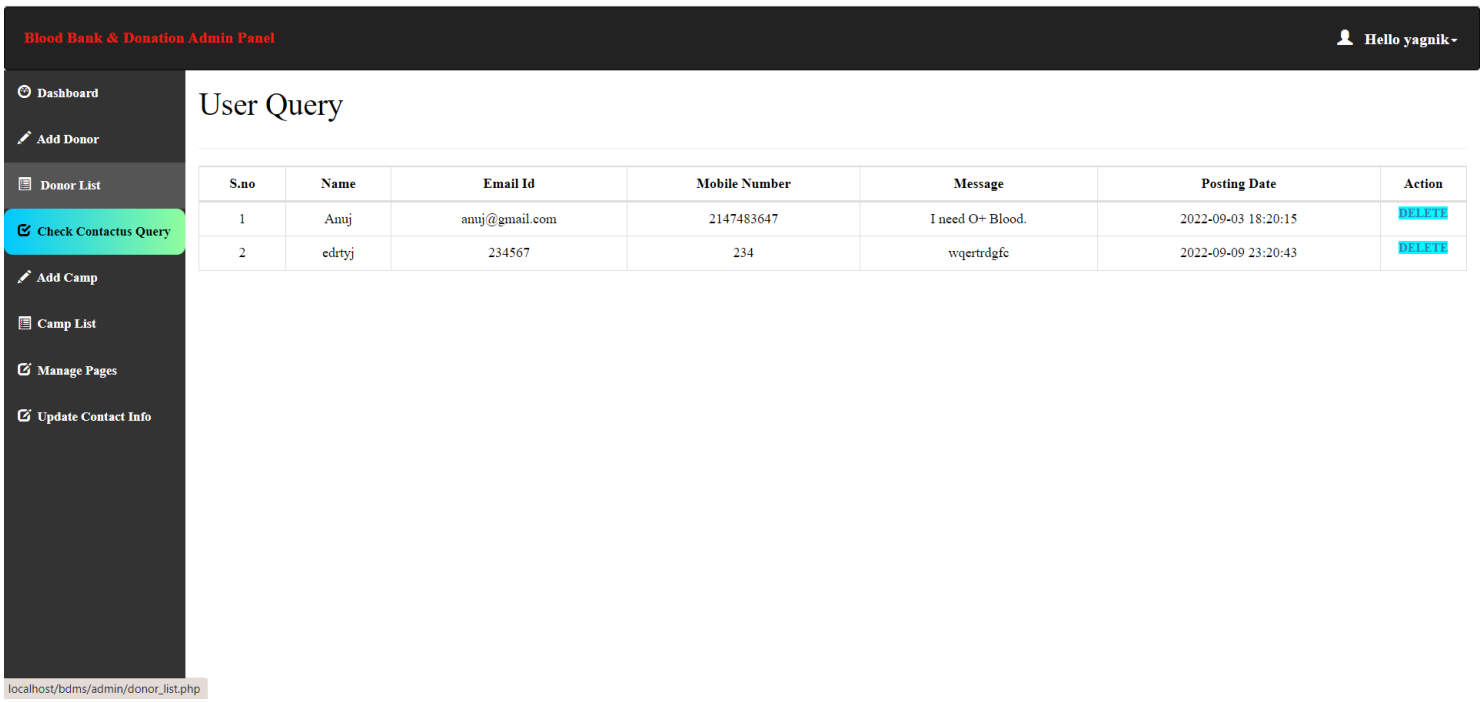
- **Add Donor Page**



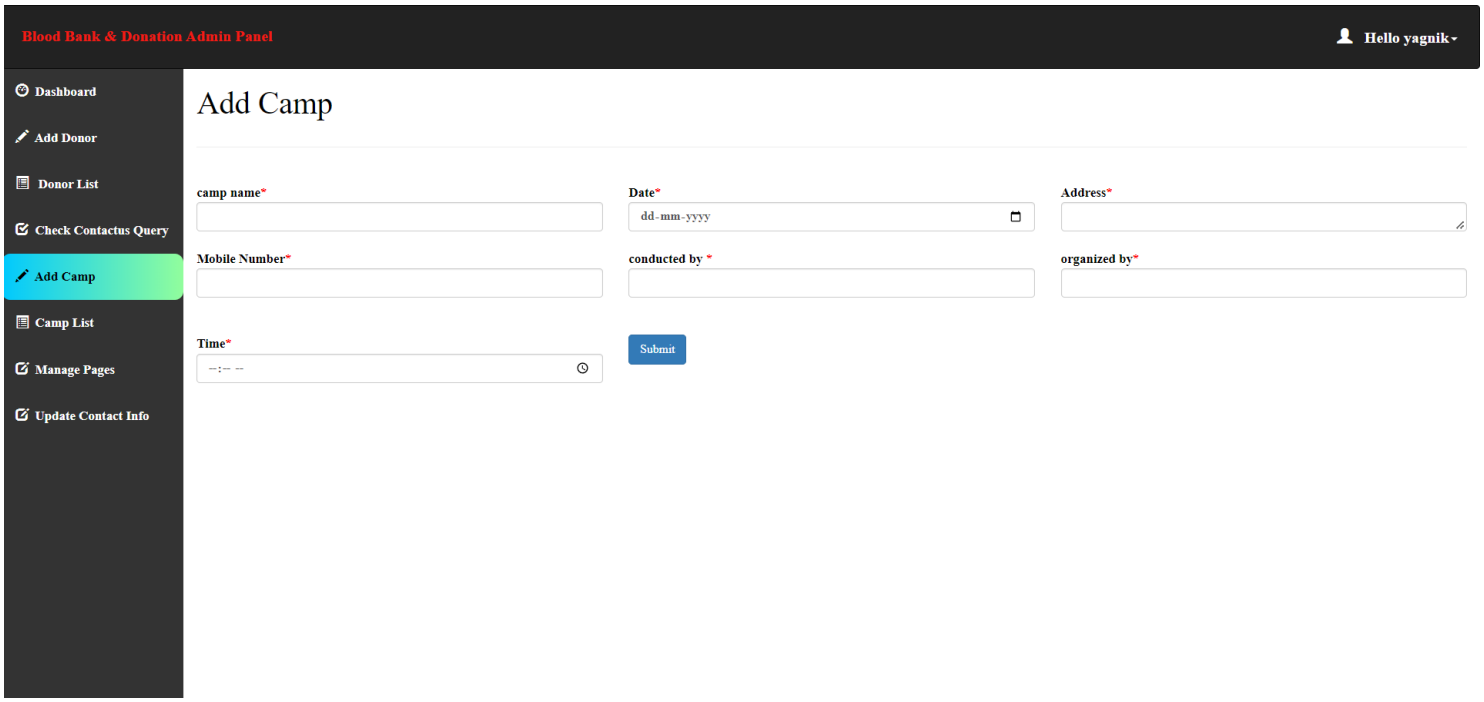
- **Donor List Page**



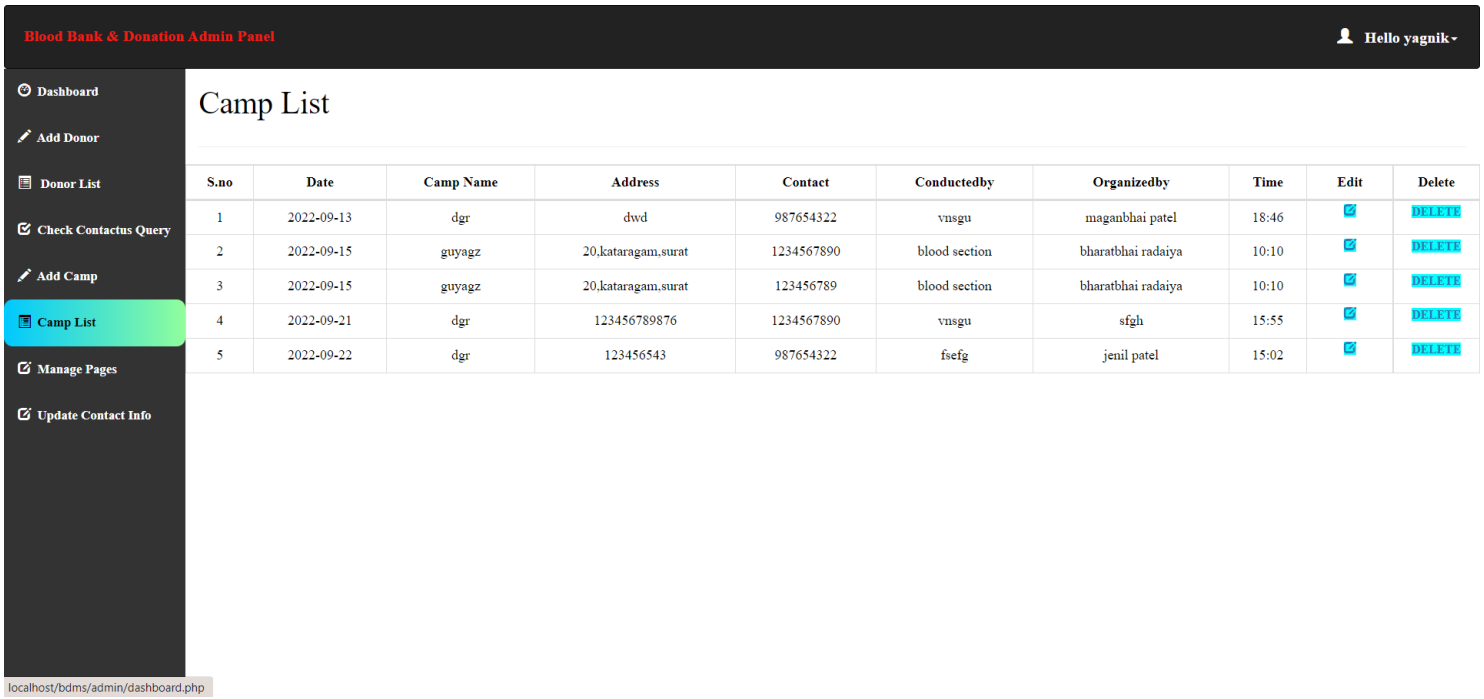
- **User Query Page**



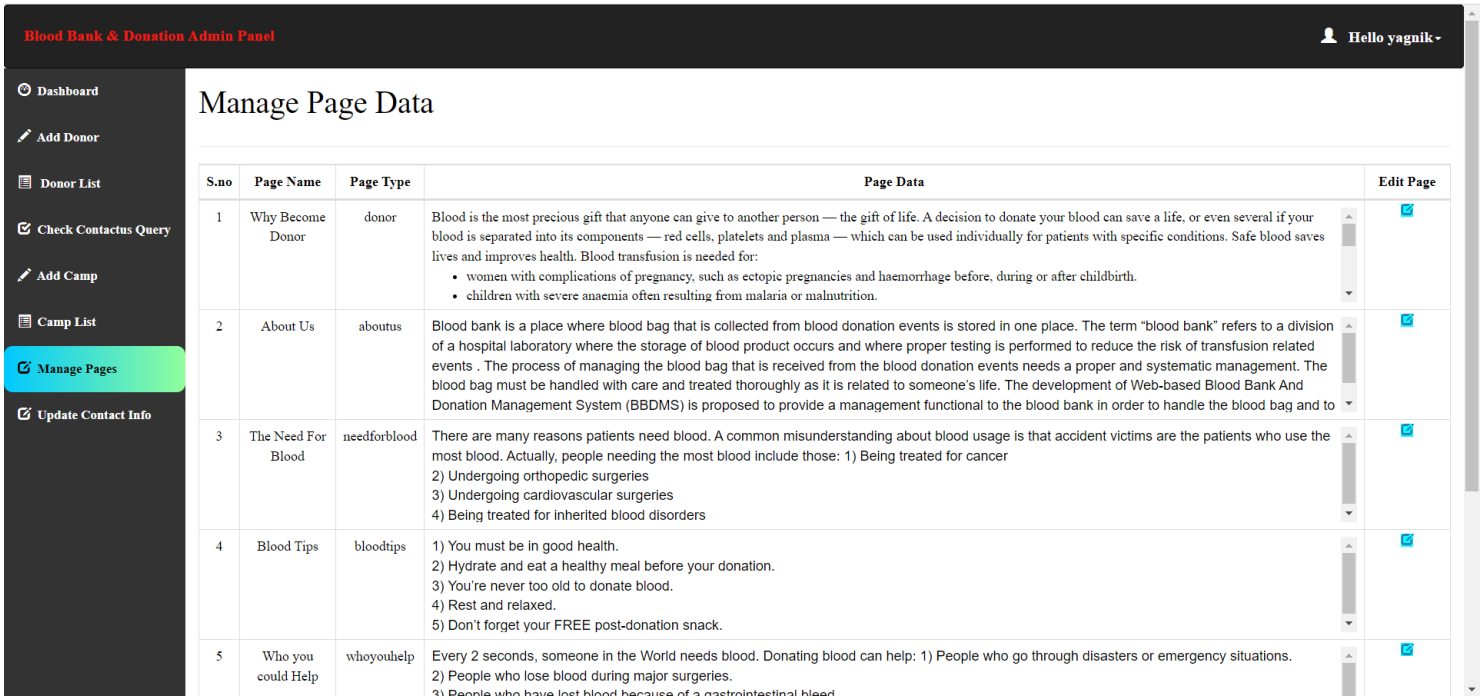
- **Add Camp Page**



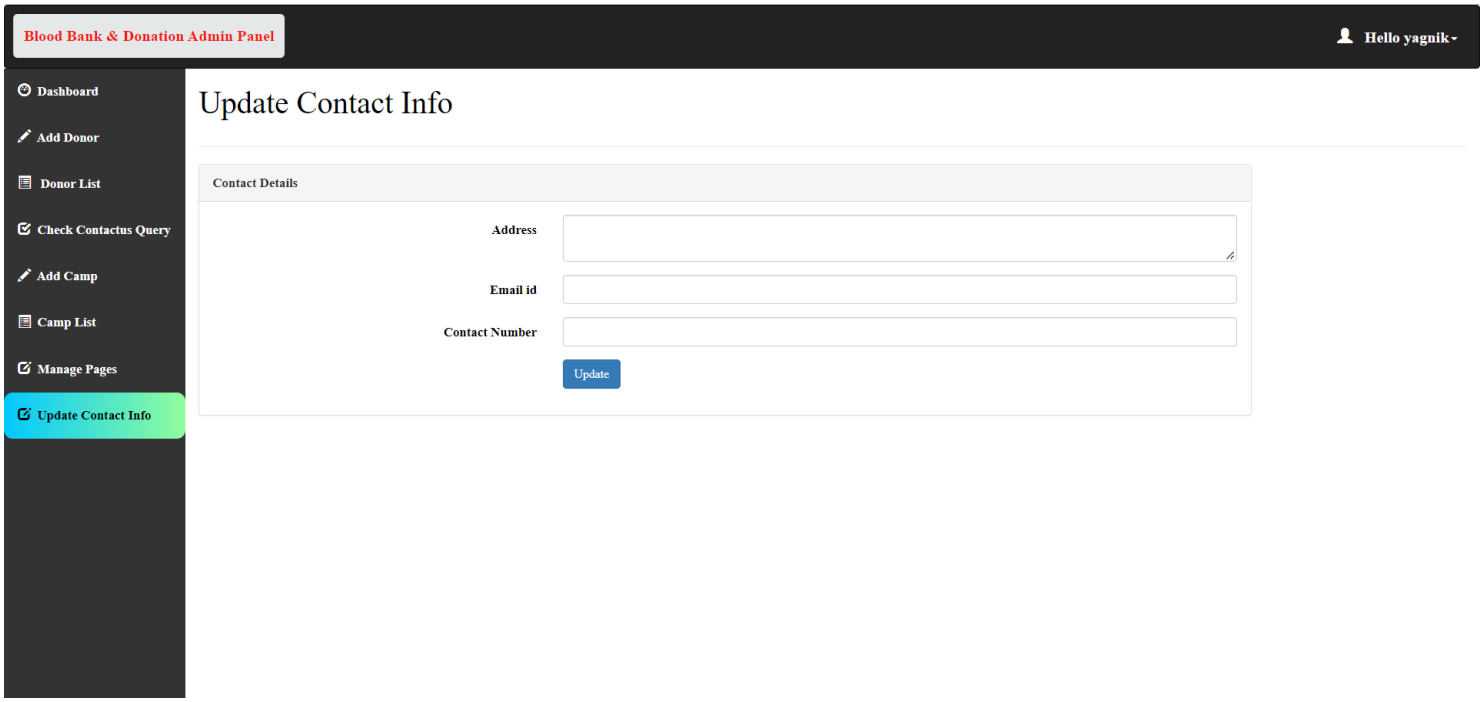
- **Camp List Page**



- **Pages Data Manage**

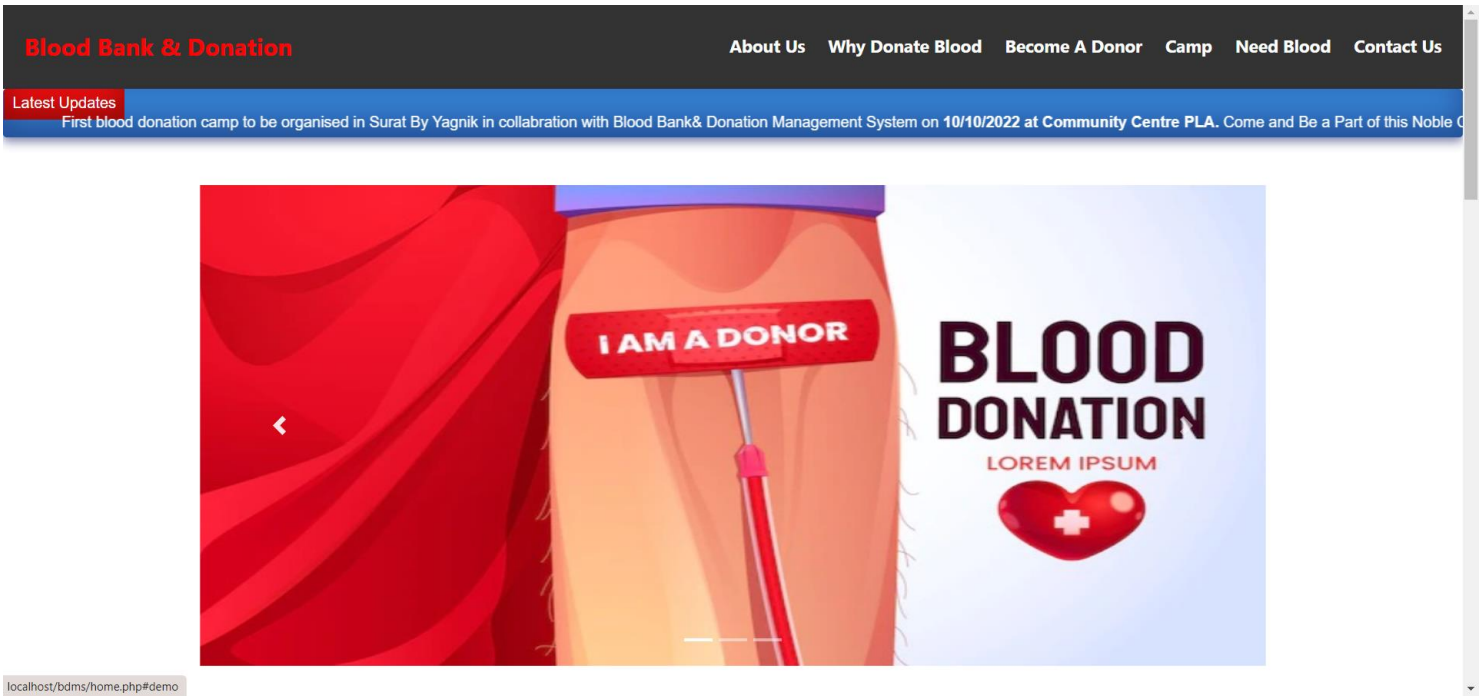


- **Update Admin Contact Info Page**



User Side :

- **Home Page**


Blood Bank & Donation
About Us
Why Donate Blood
Become A Donor
Camp
Need Blood
Contact Us
Latest Updates
First blood donation camp to be organised in Surat By Yagnik in collabration with Blood Bank& Donation Management System on 10/10/2022 at Community Centre PLA. Come and Be a Part of this Noble
I AM A DONOR
BLOOD
DONATION
LOREM IPSUM
localhost/bdms/home.php#demo

Welcome to BloodBank & Donor Management System

The need for blood

There are many reasons patients need blood. A common misunderstanding about blood usage is that accident victims are the patients who use the most

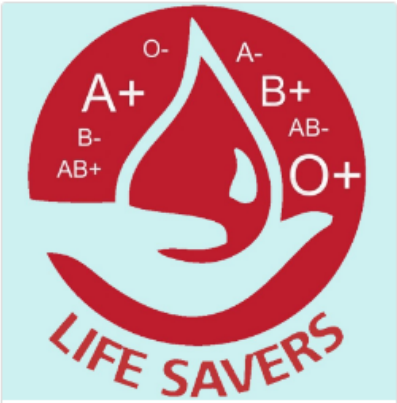
Blood Tips

3) You're never too old to donate blood.
4) Rest and relaxed.
5) Don't forget your FREE post-donation snack.

Who you could Help

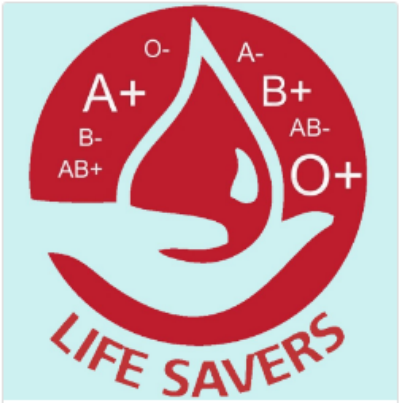
Every 2 seconds, someone in the World needs blood. Donating blood can help: 1) People who go through disasters or emergency situations.

Blood Donor Names



erthqsw

Blood Group : O+
Mobile No. : 987654321
Gender : Male
Age : 123
Address : qwertyjhbgv



erth

Blood Group : B+
Mobile No. : 2147483647
Gender : Female
Age : 18
Address : qwedfd

Camp Notice

S.no	Date	Camp Name	Address	Contact	Conductedby	Organizedby	Time
1	2022-09-13	dgr	dwd	987654322	vnsgu	maganbhai patel	18:46
2	2022-09-15	guyagz	20,kataragam,surat	1234567890	blood section	bharatbhai radaiya	10:10
3	2022-09-15	guyagz	20,kataragam,surat	123456789	blood section	bharatbhai radaiya	10:10
4	2022-09-21	dgr	123456789876	1234567890	vnsgu	sfgh	15:55
5	2022-09-22	dgr	123456543	987654322	fsefg	jenil patel	15:02

BLOOD GROUPS

Blood group of any human being will mainly fall in any one of the following groups.

- A positive or A negative
- B positive or B negative
- O positive or O negative
- AB positive or AB negative.

Your blood group is determined by the genes you inherit from your parents. A healthy diet helps ensure a successful blood donation, and also makes you feel better!



UNIVERSAL DONORS AND RECIPIENTS

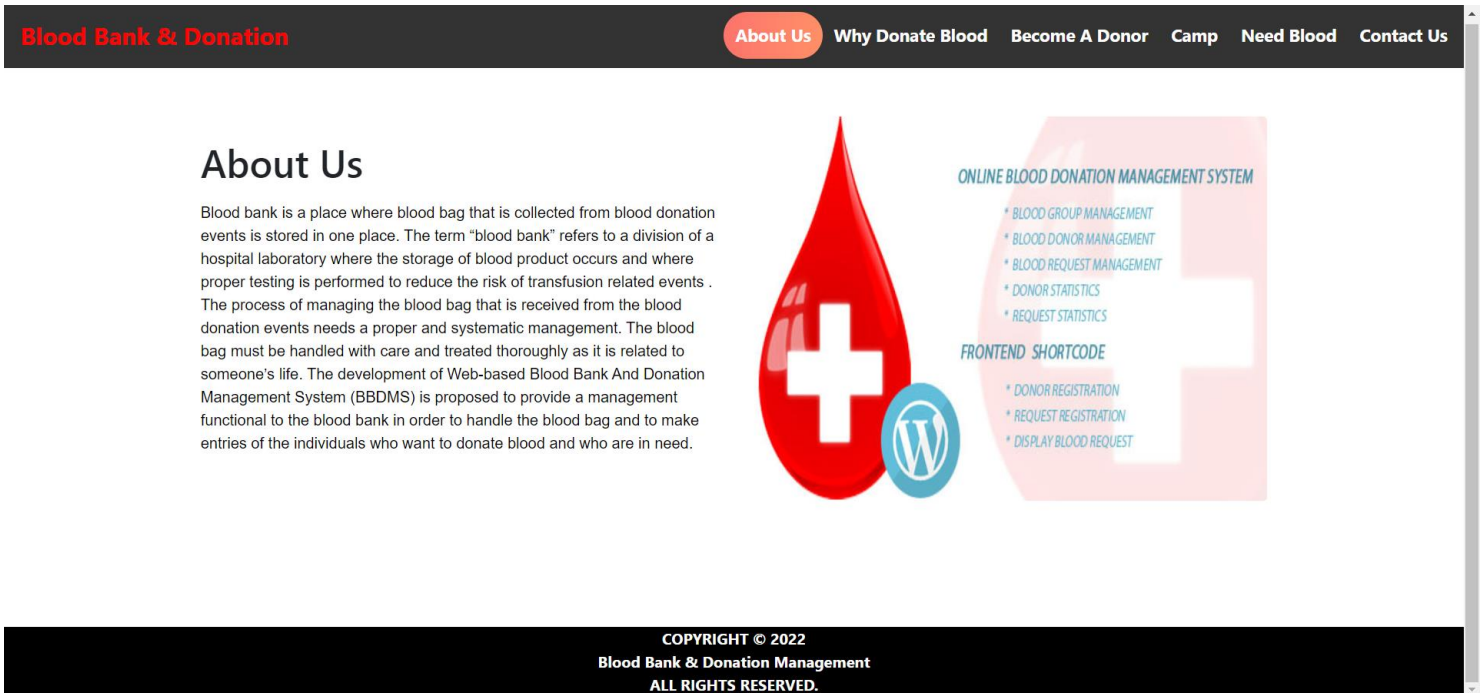
The most common blood type is O, followed by type A. Type O individuals are often called "universal donors" since their blood can be transfused into persons with any blood type. Those with type AB blood are called "universal recipients" because they can receive blood of any type.

For emergency transfusions, blood group type O negative blood is the variety of blood that has the lowest risk of causing serious reactions for most people who receive it. Because of this, it's sometimes called the universal blood donor type.

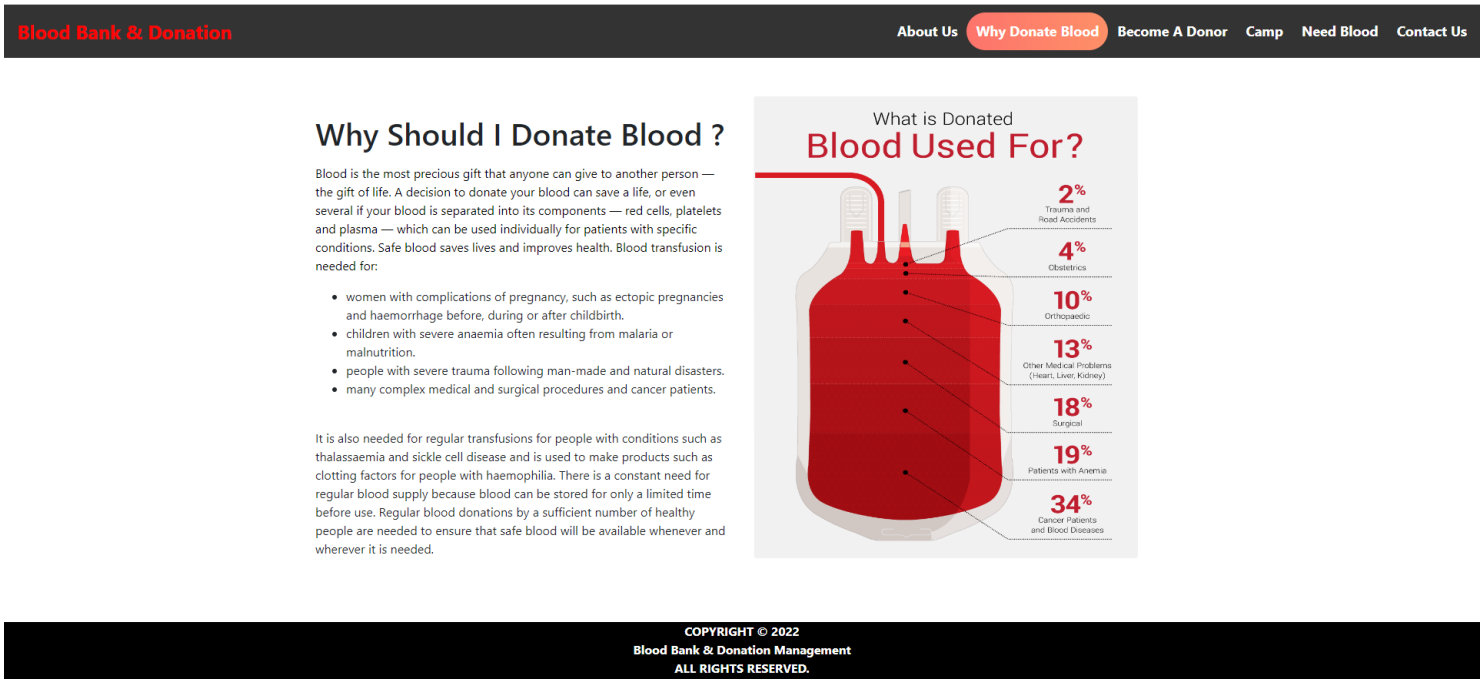
Become a Donor

COPYRIGHT © 2022
Blood Bank & Donation Management
ALL RIGHTS RESERVED.

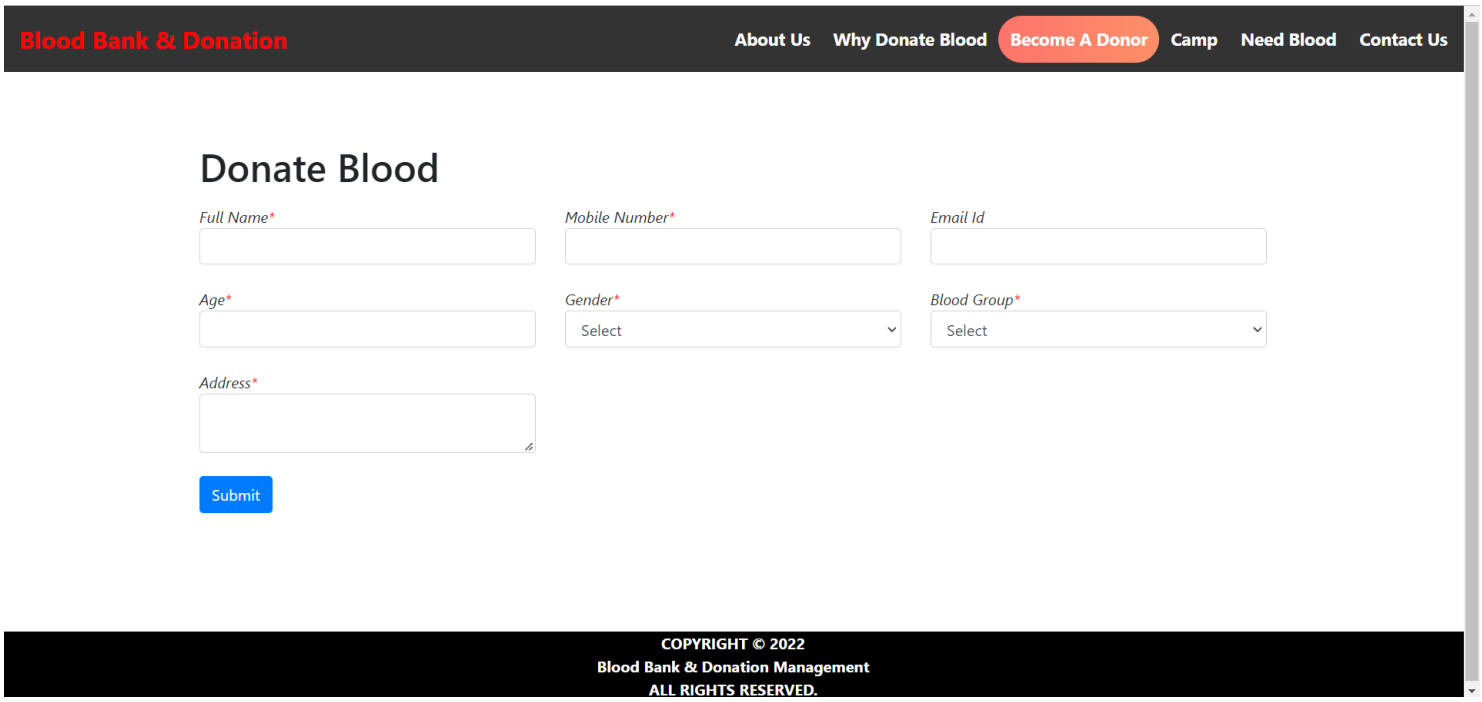
- **About Us Page**



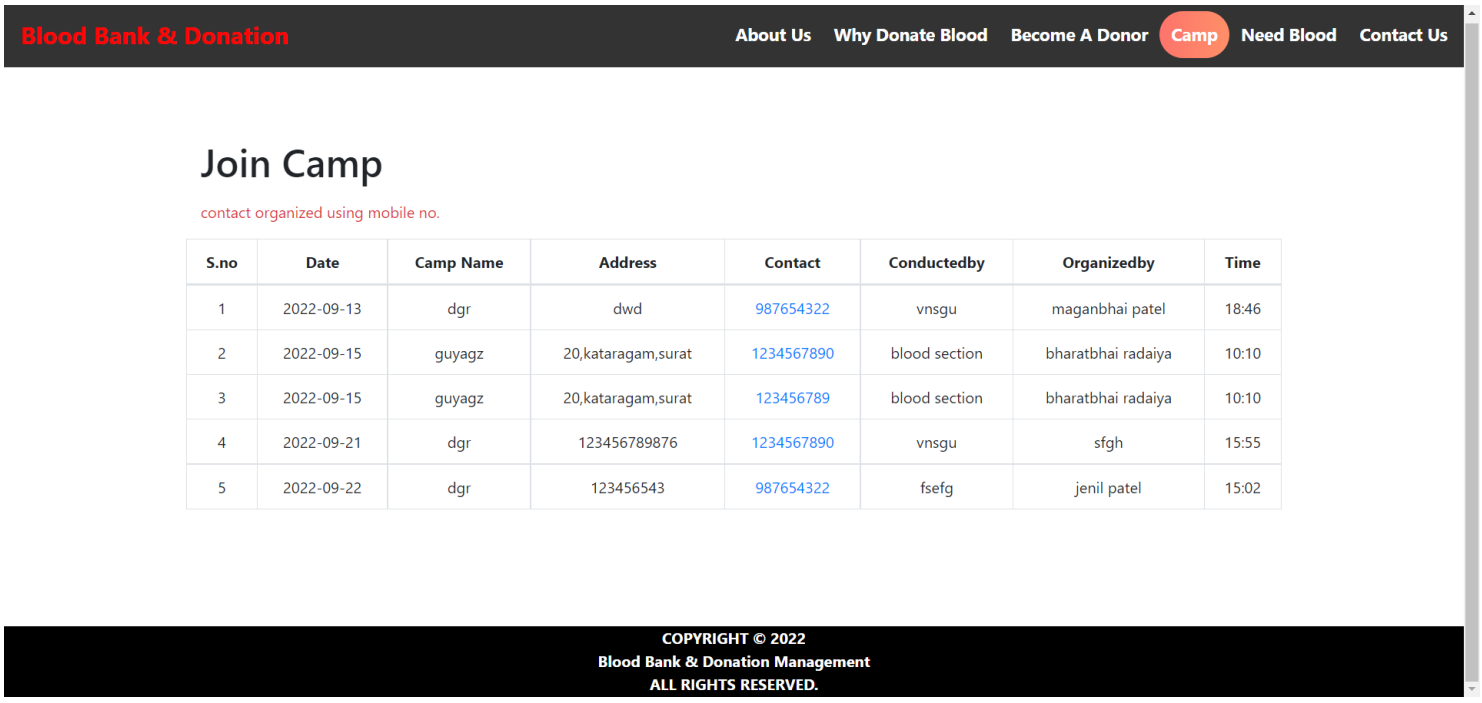
- **Why Donate Blood Page**



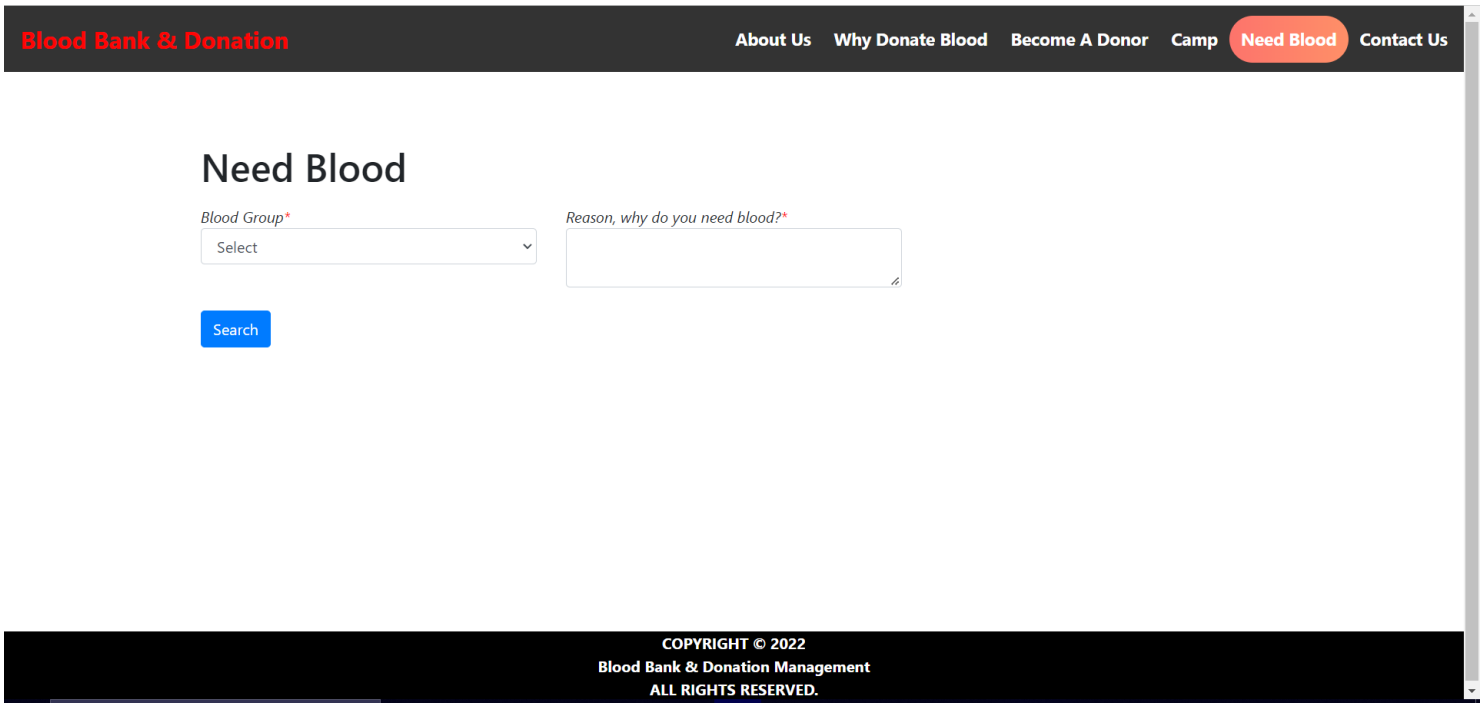
- **Donate Blood Page**



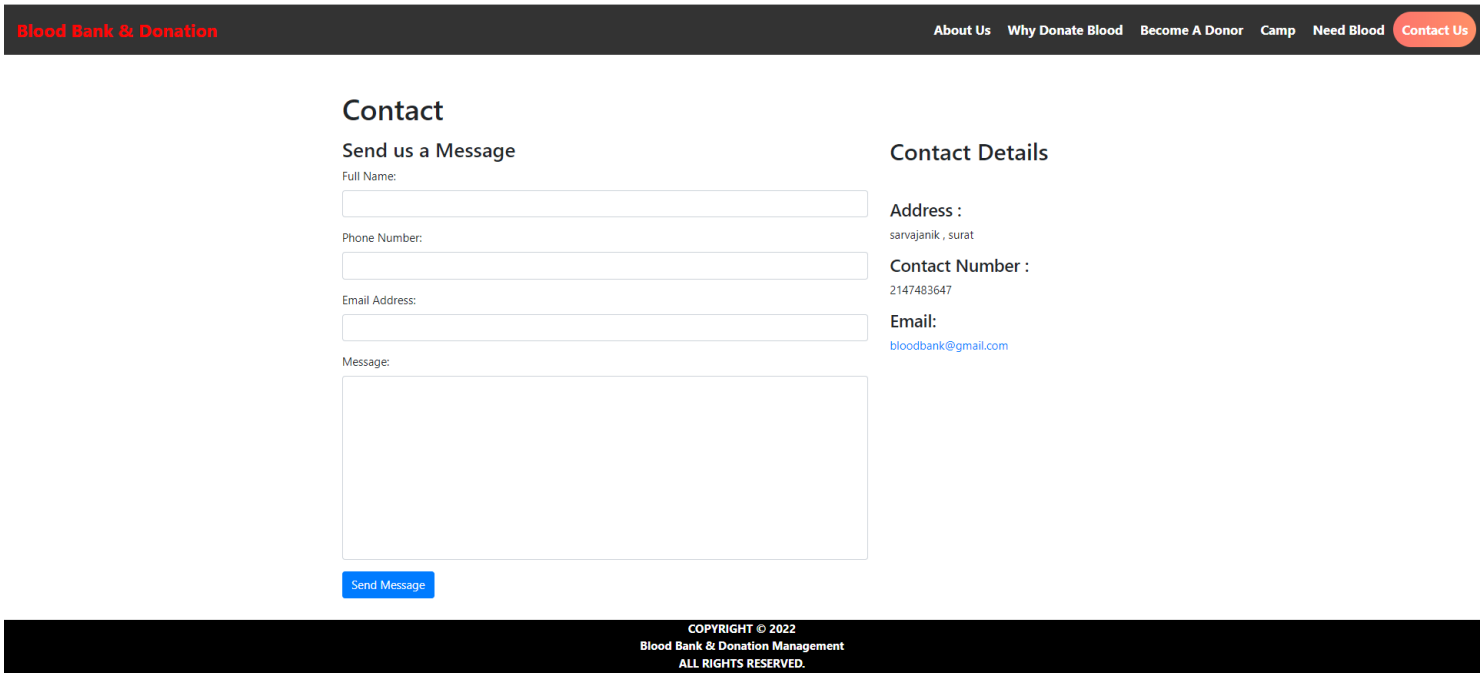
- **Camp Page**



- **Search Blood Page**



- **Admin Contact Query Page**



5. Code

User Side :

Donate Blood

```
<html>

<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">
  <meta name="description" content="">
  <meta name="author" content="">
<link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
</head>

<body>
<?php
$active ='donate';
 include('head.php') ?>

<div id="page-container" style="margin-top:50px; position: relative;min-height: 84vh;">
  <div class="container">
  <div id="content-wrap" style="padding-bottom:50px;">
<div class="row">
   <div class="col-lg-6">
      <h1 class="mt-4 mb-3">Donate Blood </h1>
     </div>
</div>
<form name="donor" action="savedata.php" method="post">
```

```
<div class="row">
<div class="col-lg-4 mb-4">
<div class="font-italic">Full Name<span style="color:red">*</span></div>
<div><input type="text" name="fullname" class="form-control" required></div>
</div>
<div class="col-lg-4 mb-4">
<div class="font-italic">Mobile Number<span style="color:red">*</span></div>
<div><input type="tel" id="phone" name="mobileno" pattern="[0-9]{10}" class="form-control" required></div>
</div>
<div class="col-lg-4 mb-4">
<div class="font-italic">Email Id</div>
<div><input type="email" name="emailid" class="form-control"></div>
</div>
</div>
<div class="row">
<div class="col-lg-4 mb-4">
<div class="font-italic">Age<span style="color:red">*</span></div>
<div><input type="number" name="age" class="form-control" min="19" required></div>
</div>
<div class="col-lg-4 mb-4">
<div class="font-italic">Gender<span style="color:red">*</span></div>
<div><select name="gender" class="form-control" required>
<option value="">Select</option>
<option value="Male">Male</option>
<option value="Female">Female</option>
</select>
</div>
</div>
<div class="col-lg-4 mb-4">
<div class="font-italic">Blood Group<span style="color:red">*</span></div>
<div><select name="blood" class="form-control" required>
```

```
<option value=""selected disabled>Select</option>
<?php
include 'conn.php';
$sql= "select * from blood";
$result=mysqli_query($conn,$sql) or die("query unsuccessful.");
while($row=mysqli_fetch_assoc($result)){
?>
<option value=" <?php echo $row['blood_id'] ?>"> <?php echo $row['blood_group'] ?> </option>
<?php } ?>
</select>
</div>
</div>
</div>
<div class="row">
<div class="col-lg-4 mb-4">
<div class="font-italic">Address<span style="color:red">*</span></div>
<div><textarea class="form-control" name="address" required></textarea></div></div>
</div>
<div class="row">
<div class="col-lg-4 mb-4">
<div><input type="submit" name="submit" class="btn btn-primary" value="Submit"
style="cursor:pointer"></div>
</div>
</div>
</div>
</div>
<?php include('footer.php') ?>
</div>
</body>
</html>
```

Need Blood

```
<html>

<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">
  <meta name="description" content="">
  <meta name="author" content="">
<link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
</head>

<body>
  <?php
  $active ='need';
  include('head.php') ?>

  <div id="page-container" style="margin-top:50px; position: relative;min-height: 84vh;">
    <div class="container">
    <div id="content-wrap" style="padding-bottom:50px;">

  <div class="row">
      <div class="col-lg-6">
          <h1 class="mt-4 mb-3">Need Blood</h1>

        </div>
  </div>
  <form name="needblood" action="" method="post">
  <div class="row">
  <div class="col-lg-4 mb-4">
```

```
<div class="font-italic">Blood Group<span style="color:red">*</span></div>
<div><select name="blood" class="form-control" required>
<option value=""selected disabled>Select</option>
<?php
include 'conn.php';
$sql= "select * from blood";
$result=mysqli_query($conn,$sql) or die("query unsuccessful.");
while($row=mysqli_fetch_assoc($result)){
?>
<option value=" <?php echo $row['blood_id'] ?>"> <?php echo $row['blood_group'] ?> </option>
<?php } ?>
</select>
</div>
</div>

<div class="col-lg-4 mb-4">
<div class="font-italic">Reason, why do you need blood?<span style="color:red">*</span></div>
<div><textarea class="form-control" name="address" required></textarea></div></div>
</div>
<div class="row">
<div class="col-lg-4 mb-4">
<div><input type="submit" name="search" class="btn btn-primary" value="Search"
style="cursor:pointer"></div>
</div>

</div><div class="row">
<?php if(isset($_POST['search'])){

$bg=$_POST['blood'];
$sql= "select * from donor_details join blood where donor_details.donor_blood=blood.blood_id AND
donor_blood='{$bg}' order by rand() limit 5";
$result=mysqli_query($conn,$sql) or die("query unsuccessful.");
```

```
   if(mysqli_num_rows($result)>0)   {
   while($row = mysqli_fetch_assoc($result)) {
     ?>


     <div class="col-lg-4 col-sm-6 portfolio-item" ><br>
     <div class="card" style="width:300px">
        <img class="card-img-top" src="image\blood_drop_logo.jpg" alt="Card image"
style="width:100%;height:300px">
        <div class="card-body">
          <h3 class="card-title"><?php echo $row['donor_name']; ?></h3>
          <p class="card-text">
            <b>Blood Group : </b> <b><?php echo $row['blood_group']; ?></b><br>
            <b>Mobile No. : </b> <?php echo $row['donor_number']; ?><br>
            <b>Gender : </b><?php echo $row['donor_gender']; ?><br>
            <b>Age : </b> <?php echo $row['donor_age']; ?><br>
            <b>Address : </b> <?php echo $row['donor_address']; ?><br>
          </p>


        </div>
       </div>
  </div>


  <?php
    }
  }
    else
    {

       echo '<div class="alert alert-danger">No Donor Found For your search Blood group </div>';


    }
} ?>
```

```
</div>
</div>
</div>
<?php include 'footer.php' ?>
</div>
</body>

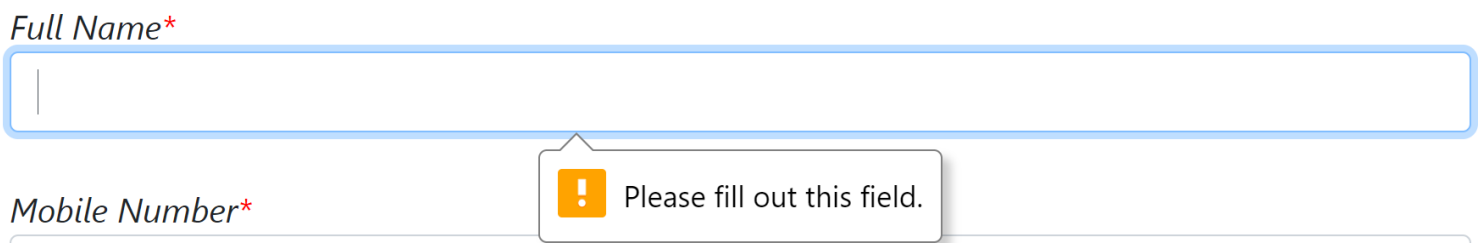
</html>
```

6. Testing

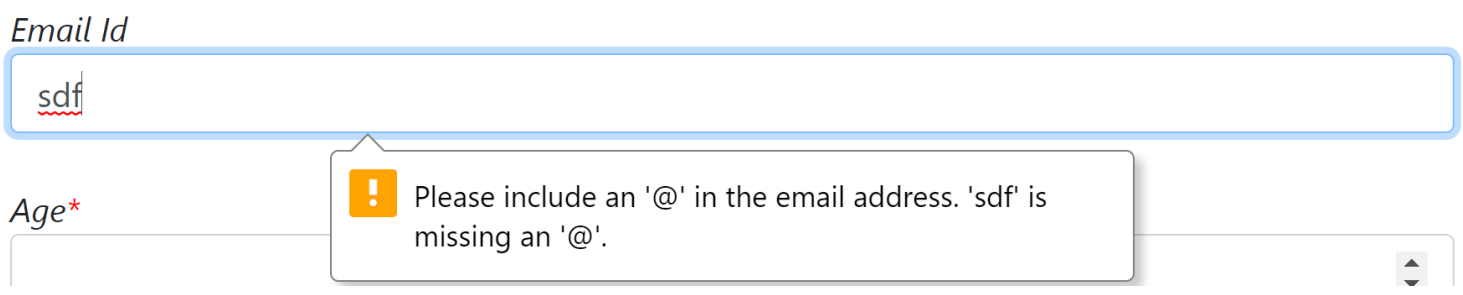
❖ Registration Page:

NO.	Input	Description	Expected Output	Actual Output	Result
1	Empty UserName	Blank	Error	Error	Registration Fail
2	Empty Email	Blank/Wrong/Not Valid Formate	Error	Error	Registration Fail
3	Empty Mobile no	Blank/ Require 10 Digit	Error	Error	Registration Fail
4	Empty /18 above age	Blank/Wrong	Error	Error	Registration Fail

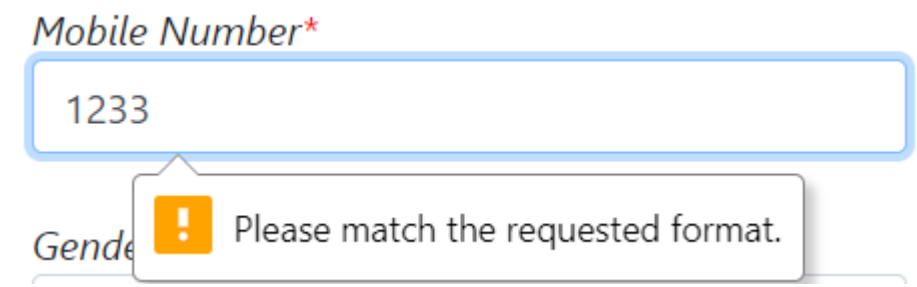
1)



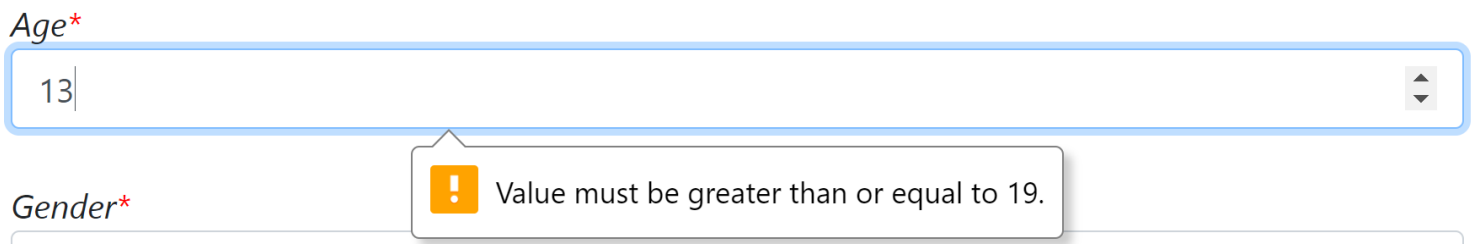
2)



3)


Mobile Number*
1233
Please match the requested format.
Gende

4)


Age*
13
Value must be greater than or equal to 19.
Gender*

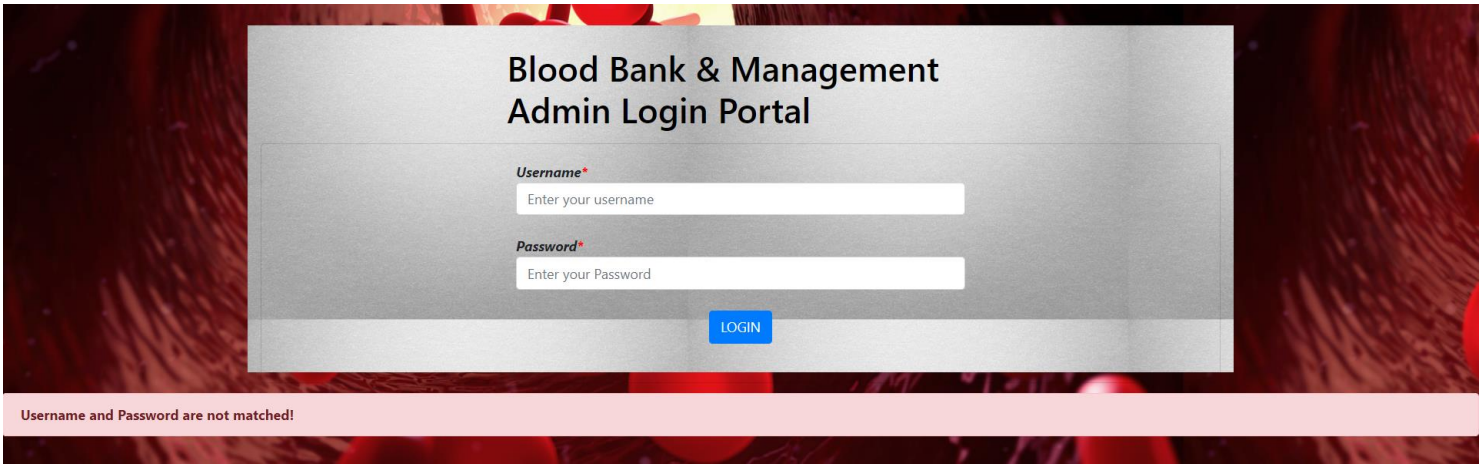
❖ Login Page:

NO.	Input	Description	Expected Output	Actual Output	Result
1	Empty UserName	Blank	Error	Error	Login Fail
2	Empty /Wrong password	Blank/Wrong	Error	Error	Login Fail

1)



2)



❖ Camp Page:

NO.	Input	Description	Expected Output	Actual Output	Result
1	Empty CampName	Blank	Error	Error	Camp Registration Fail
2	Empty Date	Blank	Error	Error	Camp Registration Fail
3	Empty Mobile no	Blank Require 10 Digit	Error	Error	Camp Registration Fail
4	Empty /Address	Blank	Error	Error	Camp Registration Fail
5	Empty /Time	Blank	Error	Error	Camp Registration Fail
6	Empty /conducted /organized	Blank	Error	Error	Camp Registration Fail

camp name*

Date*

dd-mm-yyyy

Address*

Mobile Number*

conducted by *

organized by*

Time*

--:-- --

Submit

7.0 NEW TOOLS/ TECHNOLOGIES LEARNED/ USED:

- HTML
- CSS
- JavaScript
- Bootstrap
- PHP
- My SQL Database (PHP My Admin)

•

8.0 SYSTEM LIMITATION/CONSTRAINTS:

- User can not get email
- Admin not add directly admin add by other admin
- Camp contact using website

9.Future Enhancement & Opportunities

- More efficient design and user experience.
- Apply More efficient constraints .
- Add module to admin can change password using Mobile No./Email OTP.
- Add feature of login/signup using google / facebook.

10. BIBLIOGRAPHY & REFERENCES:

Books

- PHP: Learn PHP in 24 Hours or Less - A Beginner's Guide To Learning PHP Programming Now

Website

- www.W3School.com
- www.youtube.com